

1. Batch G.D.
→ uses all training samples before updating weights.

Advantages
→ stable convergence
→ Accurate gradient calculation

Disadvantages
→ slow for large dataset
→ requires more memory

2. Stochastic G.D

The model updates its weight after processing each individual training sample, rather than waiting for entire dataset.

3. Mini-batch G.D.

Instead of using one sample or the entire dataset, Mini-batch G.D. uses a small batch of samples.

Ex: Dataset: 100 images

Batch size = 100 images

Weight updates:

100 images → update.

Next 100 images → update

Next 10 images → update

16/7/2020

Curse of Dimensionality:

→ The curse of dimensionality refers to the challenges that arise when working with high-dimensional data, where increasing the number of features leads to reduced model performance and higher computational cost. Computation complexity increases for the

→ It becomes more difficult for the model to identify meaningful patterns.

How to overcome the curse of dimensionality

1) Feature Selection (Choose relevant features)

Ex:

for student performance prediction

- × Study hours
- × Attendance
- × Previous marks

2) Use Appropriate Algorithms

Ex: → Random Forest

- XGBoost
- CNN
- Deep Neural Networks

3) Dimensionality Reduction

→ Principal Component Analysis (PCA)

→ Linear Discriminant Analysis (LDA)

→ Auto encoder in deep learning

3) 1143

$$\begin{bmatrix} 4 & 5 \\ 7 & 8 \end{bmatrix} = (4 \times 1) + (5 \times 0) + (7 \times 0) + (8 \times 1) \\ = 4 + 8 \\ = 12$$

$$4) \begin{bmatrix} 5 & 6 \\ 8 & 9 \end{bmatrix} = (5 \times 1) + (6 \times 0) + (8 \times 0) + (9 \times 1) \\ = 5 + 9 \\ = 14$$

final feature map:

$$\begin{bmatrix} 12 & 8 \\ 12 & 14 \end{bmatrix}$$

Input

$$\begin{bmatrix} 3 & 2 & 1 \\ 1 & 0 & 2 \\ 2 & 1 & 3 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 3 & 2 \\ 1 & 0 \end{bmatrix}$$

$$\Rightarrow 3 \times 0 + 1 \times 1 + 0 = 3$$

$$\Rightarrow 2 + 1 = 3$$

$$\begin{bmatrix} 3 & 1 \\ 2 & 3 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$$

$$\Rightarrow 2 \times 0 + 1 \times 1 + 0 + 1 = 2$$

$$= 0 + 1 + 1 = 2$$

$$\Rightarrow \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$$

$$= 1 \times 0 + 2 \times 1 + 0 + 1 = 3$$

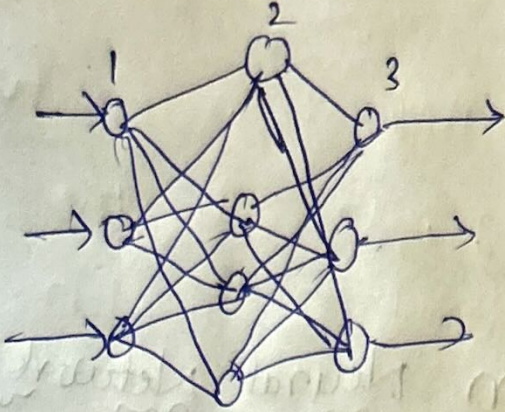
$$\Rightarrow \begin{bmatrix} 0 & 2 \\ 1 & 3 \end{bmatrix}$$

$$= 3 \times 0 + 2 \times 1 + 1 \times 1 + 3 \times 0 = 0 + 2 + 1 + 0 = 3$$

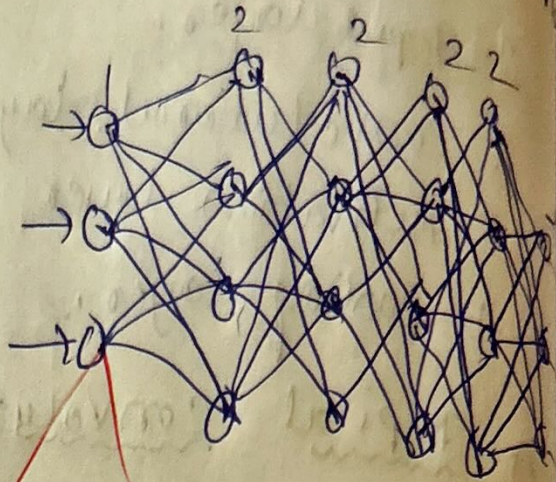
$$= 2$$

25/12/20

Neural Networks



Deep Neural Network



1. Input

2. Hidden layer

3. output layer

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